Real-Time Medical Asset Tracking System

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Introduction

Organization

Cook Children's Health Care System is a leading pediatric health network delivering complex, high-demand patient care across multiple clinical facilities.

Challenge

Hospital Nurses spend up to 60 minutes per shift searching for mobile medical equipment, leading to clinical inefficiencies and increased hospital operational costs (24x7, 2024).

Areas for Improvement

Eliminating manual asset tracking, reducing equipment loss rates, and optimizing asset utilization efficiency.

Proposed Solution

Implement an automated real-time medical asset tracking system powered by an Ultra-Wideband (UWB) IoT ecosystem to provide continuous, high-precision asset tracking.

Benefit

Redirects up to 60 minutes of daily nurse search time back to bedside care while eliminating redundant equipment and labor expenses.

Executive Summary

In modern healthcare operations, rapid access to mobile medical equipment is essential for direct patient care and clinical efficiency. This project focuses on eliminating time nurses spend searching for equipment across hospital units at Cook Children’s by evaluating and implementing:

- Real-time asset tracking
- Automated workflow integration that eliminates manual check-outs
- Continuous asset visibility via interactive dashboards
- Enterprise security & operational sustainability

This automated IoT solution reclaims up to 60 minutes per nurse per shift, redistributing labor hours back to bedside pediatric patient care while preventing redundant equipment purchases and loss.

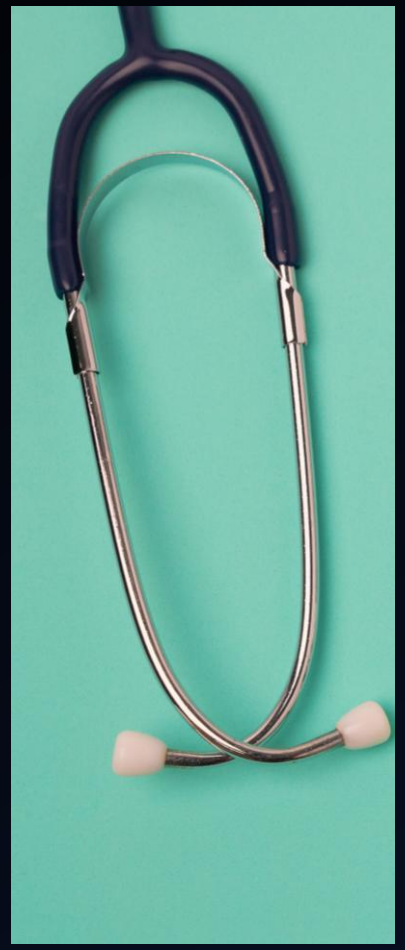
Organization Profile and Problem Statement

The Organization



Cook Children's is a top-ranked, non-profit pediatric healthcare network headquartered in Fort Worth, Texas. Founded with a commitment to improving the health and well-being of children, the organization has grown into a comprehensive healthcare system serving patients and families throughout North Texas. Cook Children's combines specialized pediatric medical care with research, education, and community health initiatives to address the diverse healthcare needs of children and their families. As a pediatric organization, Cook Children's prioritizes patient care and works to provide accessible, high-quality healthcare across its network.

Products and Services



Medical Care

- Primary and specialty pediatric care
- Emergency and urgent care
- Inpatient and outpatient treatment
- Pediatric surgery
- Rehabilitation
- Behavioral health

Support Services

- Diagnostic testing
- Pharmacy services
- Care coordination
- Home health services
- Health education

Specialized and Community Services

- Complex-care programs
- Pediatric medical centers
- Specialty programs
- Community health initiatives

Areas for Improvement with Technology

Medical Equipment Management and Tracking



Concerns

- Nurses spend valuable time searching for shared medical equipment (Qtait, 2025).
- Equipment may be in another room, department, or floor.
- Nurses may not know whether equipment is available, clean, or ready for use.
- Time spent searching takes nurses away from direct patient care.

Improvements

- Attach an IoT sensor to mobile medical equipment.
- Provide real-time equipment location and availability.
- Nurses badges automatically record equipment checkout and usage.
- Track cleaning and maintenance status.
- Provide a dashboard for managing equipment.

Benefit

- Reduce the time nurses spend searching for equipment and give that time back to patient care.
- Real-time asset tracking research has demonstrated potential for improving the efficiency of locating hospital equipment (Qtait, 2025).

Problem Statement



Nurses spend significant amounts of time manually tracking medical equipment, which is an operational inefficiency that disrupts clinical workflows, reduces direct patient care time, and can contribute to unnecessary spendings (Qtait, 2025).

Technology Selection and Design

Alternative Technology Approaches

Checkout System

Hardware
Wall-mounted kiosks with RF scanners + Passive RFID asset tags

Network
Logs transactions on a local DB over an isolated encrypted hospital LAN

Workflow
Two-step scan (Staff Badge + Asset Tag)

Software
Windows GUI (Inventory count, active checkouts, logs)

Access Control
- Nurses: Read availability / write check-ins & transfers
- Charge Nurses: Add assets, clear forgotten checkouts, mark missing or damaged assets
- Admins: Full permission & log management

IoT Ecosystem

Hardware
Ceiling gateways + Active Ultra-Wideband (UWB) asset tags (with Clean, Dirty, and Maintenance buttons)

Network
Encrypted Wi-Fi VLAN streaming to local DB

Workflow
Automated room-level tracking that pairs UWB assets with nurse room assignments (Zero manual scanning)

Software
Real-time visual floor map, inventory alerts, geofencing to alert if equipment leaves the facility

Access Control:
- Nurses: View live asset maps & toggle hygiene states
- Charge Nurses: Manage stock thresholds & maintenance escalations
- Admins: Gateway, network, and permission configuration

Qualitative Analysis

Checkout System

Operation: Intermittent
Requires physical stops at kiosks to scan badges & equipment tags.

Availability: Dependent on Infrastructure
Relies on continuous kiosk uptime, local LAN, and database connectivity.

Cost: High
Driven by kiosk hardware and ongoing labor overhead.

Implementation: Simple
Modular wall mounting with standard local database setup and minimal IT friction.

Benefit to Company: Minimal
Basic audit trails prevent theft/hoarding but offers minimal real-time workflow relief.

IoT Ecosystem

Operation: Continuous
Automated tracking requires zero staff scan time and eliminates manual check-out stops.

Availability: Continuous Asset Visibility
Segmented Wi-Fi VLAN provides real-time, room-level tracking.

Cost: Moderate
Gateway and tag investments are offset by substantial long-term labor savings.

Implementation: Complex
Requires ceiling gateway installation, Wi-Fi VLAN segmentation, and tag calibration.

Benefit to Company: Enterprise Efficiency
Drastically reduces nurse search times, automates restocking, and prevents asset loss.

Quantitative Analysis

Checkout System

Operation: Intermittent Workflow Kiosk Burden
Consumes unnecessary time during two scanning transaction (badge + tag)

Availability: Restricted
Visibility limited to last scan, with 0% accuracy once moved from kiosk.

Cost: High
Passive RFID tags average $8 a unit and wall kiosks average $1,200–$1,800/station (Halstead, 2020).

Implementation: Rapid Local Deployment
0 hours of RF map calibration needed (Halstead, 2020).

Benefit to Company: Reduces Asset Loss
Cuts current asset loss of 15% (Altus) but offers minimal real-time relief for daily nurse searches.

IoT Ecosystem

Operation: Automated Zero-Touch
Precise tracking requiring 0 staff scan time; reclaims 30–60 min/shift (Margarita, 2026; Collins, 2025).

Availability: Continuous
Real-time visibility across all active hospital zones.

Cost: Moderate
Real-time tags cost $15–$50 each and ceiling gateways cost $150–$400 each (CPCON Group, 2026).

Implementation: Complex
Requires ceiling gateway deployment & 3–5-year battery cycle management (CPCON Group, 2026).

Benefit to Company: Enterprise ROI
Reclaims portion of $14B industry productivity loss (Margarita, 2026) and increases asset utilization from 35% to 70% (Jina, 2024).

Recommended Solution: IoT Ecosystem

Infrastructure Architecture

IoT Sensors

- UWB tags send real-time signals to ceiling gateways.
- Integrated physical buttons: Clean / Dirty / Maintenance

Ceiling Gateways

Transmits location data over isolated, encrypted Wi-Fi VLAN to a hybrid Relational/NoSQL DB.

Automated Workflow

Zero-touch room-level tracking that pairs UWB asset tags directly with nurse room assignments upon entry

Software & Access Control

User Dashboard

- Live visual floor maps tracking exact asset positions.
- Low-stock inventory alerts with charge nurse escalation thresholds.

Role-Based Access Control

- Nurses: View floor maps, check availability, toggle hygiene states.
- Charge Nurses: Reallocate stock, manage maintenance, clear alerts.
- System Administrators: Manage RBAC rules, gateway states, and network configurations.

Justification & Impact

Clinical Productivity

Reclaims 30–60 min/shift per nurse by replacing manual check-outs with room proximity pairing (Margarita, 2026; Collins, 2025).

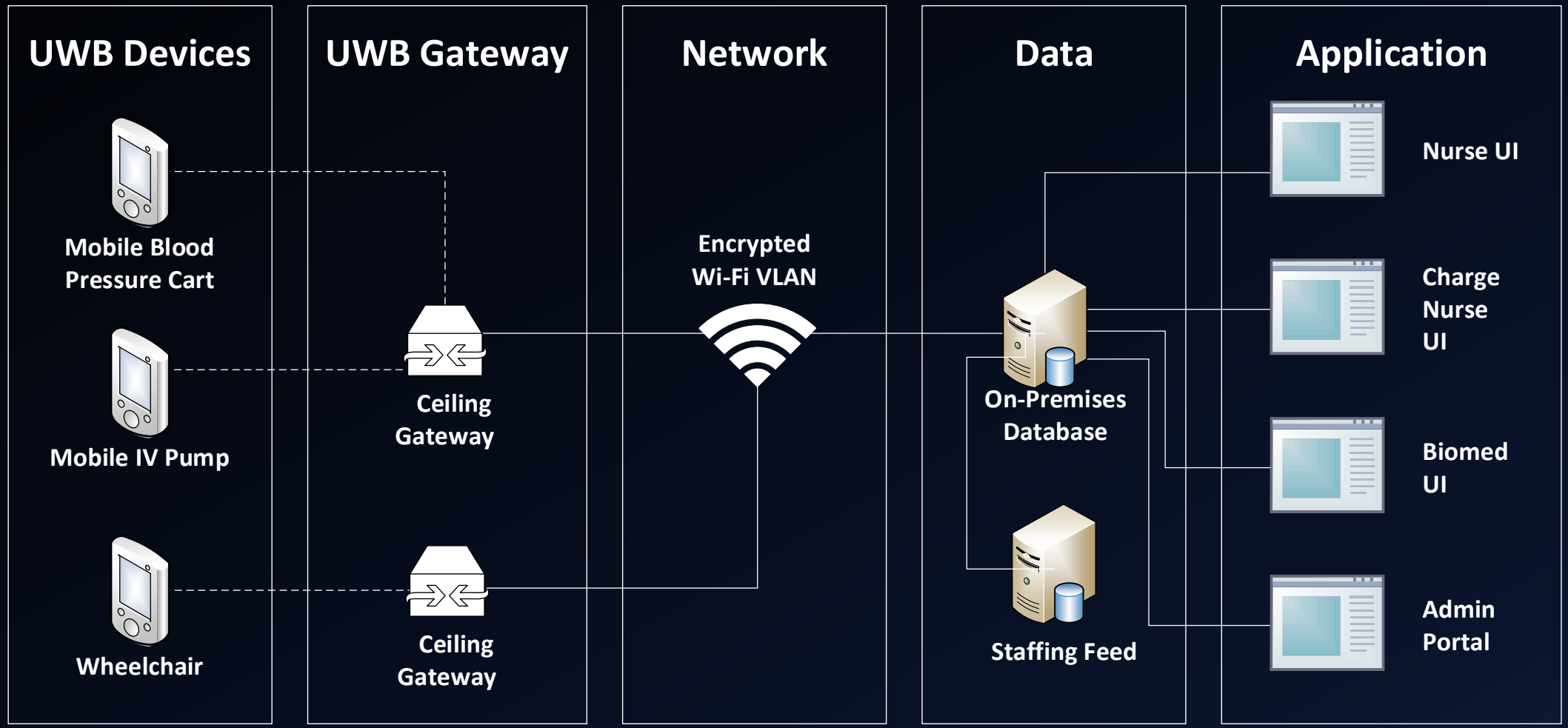
Cost & Asset Efficiency

- Eliminates redundant equipment purchases and cuts loss via automated geofencing.
- Redirects wasted search hours directly back to patient care (Margarita, 2026).

Cost & Asset Efficiency

Segmented Wi-Fi VLAN isolates tracking data, minimizing PHI exposure while integrating with staff assignment feeds.

High-Level Design: IoT Ecosystem



Implementation Plan

Work Breakdown Structure

WBS Number	Phase	Description	Owner
1	Initiation, Planning & Requirements Analysis		
1.1	Project Charter & Scope Definition	Outline project goals, set clinical boundaries for Cook Children's, and secure stakeholder alignment.	Solutions Architect & Technical Manager
1.2	Business Requirements Document	Document key operational goals, nurse search time targets, and clinical success metrics.	Business Analyst
1.3	Cost Estimates	Estimate total expenditure across UWB tags, ceiling gateways, network setup, software licensing & labor.	Solutions Architect & Technical Manager
1.4	Risk Management & Disaster Recovery Plans	Identify operational risks, fallback procedures, and data recovery strategies for hospital systems.	Solutions Architect & Technical Manager
2	Technology Selection & Architectural Design		
2.1	Technical Requirements & Feasibility	Assess hardware specs, signal range, and physical compatibility with existing hospital infrastructure.	Systems Engineer
2.2	Software Requirements Specifications	Define dashboard feature expectations, user role rules, and equipment tracking behavior.	Lead Software Architect
2.3	High-Level Architecture Design	Map out the end-to-end data pipeline connecting active UWB tags, gateways, local servers, and user interfaces.	Solutions Architect
2.4	Software, Hardware & Network Selection	Evaluate vendor options and finalize selections.	Solutions Architect & Network Lead

WBS Number	Phase	Description	Owner
3	Development & Infrastructure Integration		
3.1	UI/UX Design	Create intuitive screen mockups for nurses, charge nurses, and systems administrators.	UI/UX Designer
3.2	UWB installation and calibration on a few devices for development	Tag a small test batch of equipment to verify signal accuracy and test hardware buttons in a lab setup.	Field Techs & IT Support
3.3	Gateway Installation	Mount UWB ceiling gateways across designated unit zones to ensure seamless room-level coverage.	Network Infrastructure Team
3.4	Network Configuration	Set up an isolated, encrypted Wi-Fi VLAN to securely transmit location data.	Network Infrastructure Team
3	Development & Infrastructure Integration		
3.5	Database Configuration	Structure the on-premises database to handle real-time data updates.	Database Administrator
3.6	Application Development	Build the core software services, live map feeds, threshold alert triggers, user login, RBAC controls, database integration, UWB device tracking, and user interfaces.	Software Engineering Team
3.7	UWB installation and calibration on all devices	Attach active UWB tags to all mobile equipment on the target floor and fine-tune room positioning.	Field Techs & IT Support

Work Breakdown Structure (Cont.)

WBS Number	Phase	Description	Owner
4	Testing & Training		
4.1	Unit Testing	Test individual software modules, database queries, and tag signal listeners to fix bugs early.	Software Engineering Team
4.2	System Integration Testing & QA	Verify that hardware, network VLANs, database processing, and live dashboards communicate seamlessly.	QA Lead
4.3	User Acceptance Testing	Conduct hands-on walkthroughs with clinical leads to confirm the system meets real unit workflow needs.	Nurse Lead & QA Lead
4.4	Clinical Staff Training & Onboarding	Train nurses and charge nurses on dashboard navigation, equipment status buttons, and floor maps. Train IT administrators on dashboard controls for RBAC, gateway settings, and network configuration.	Clinical Educator & Nurse Lead

WBS Number	Phase	Description	Owner
5	Deployment		
5.1	Release pilot into production	Launch the system on the designated inpatient unit for active daily clinical use.	Solutions Architect & DevOps Lead
5.2	Post Implementation Review	Review initial operational performance, goals, gather nurse feedback, and measure early search-time reductions.	Solutions Architect & Technical Manager
6	Maintenance		
6.1	System Handover to IT Operations	Formally transition system administration, user access management, and maintenance responsibilities.	IT Operations Lead
6.2	UWB Tag Management	Swap UWB tag batteries (3–5 year lifecycle), attach tags to new assets, track tag lifespan, and troubleshoot physical hardware issues.	Biomedical Technicians
6	Maintenance		
6.3	VLAN & Security Audits	Periodically audit the isolated, encrypted Wi-Fi VLAN to ensure network stability and zero exposure to Electronic Health Records (EHR).	IT Operations
6.4	Hardware & Gateway Updates	Firmware patches and security updates to ceiling gateways and network hardware.	IT Operations
6.5	Calibration Management	Conduct annual precision re-calibrations of ceiling gateways.	IT Operations / Vendor
6.6	Application Management	Update, patch, and maintain user dashboard interfaces, database endpoints, and threshold alerts.	DevOps

Project Schedule

Task & Owner		Start Date	Due Date	Days	Week 1					Week 2					Week 3					Week 4				
					M	T	W	T	F	M	T	W	T	F	M	T	W	T	F	M	T	W	T	F
Initiation, Planning & Requirements Analysis		2/1/2027	2/26/2027	20																				
Project Charter & Scope Definition	Solutions Architect & Technical Manager	2/1/2027	2/5/2027	5																				
Business Requirements Document	Business Analyst	2/8/2027	2/12/2027	5																				
Cost Estimates	Solutions Architect & Technical Manager	2/15/2027	2/19/2027	5																				
Risk Management & Disaster Recovery Plans	Solutions Architect & Technical Manager	2/22/2027	2/26/2027	5																				

Task & Owner		Start Date	Due Date	Days	Week 5					Week 6					Week 7					Week 8				
					M	T	W	T	F	M	T	W	T	F	M	T	W	T	F	M	T	W	T	F
Technology Selection & Architectural Design		3/1/2027	3/26/2027	20																				
Technical Requirements & Feasibility	Systems Engineer	3/1/2027	3/5/2027	5																				
Software Requirements Specifications	Lead Software Architect	3/8/2027	3/12/2027	5																				
High-Level Architecture Design	Solutions Architect	3/15/2027	3/19/2027	5																				
Software, Hardware & Network Selection	Solutions Architect & Network Lead	3/22/2027	3/26/2027	5																				

Task & Owner		Start Date	Due Date	Days	Week 8					Week 9					Week 10					Week 11				
					M	T	W	T	F	M	T	W	T	F	M	T	W	T	F	M	T	W	T	F
Development & Infrastructure Integration		3/22/2027	5/28/2027	50																				
UI/UX Design	UI/UX Designer	3/22/2027	3/26/2027	5																				
UWB installation and calibration on a few devices for development	Field Techs & IT Support	3/29/2027	4/2/2027	5																				
Gateway Installation	Network Infrastructure Team	4/5/2027	4/16/2027	10																				

Project Schedule (Cont.)

Task & Owner		Start Date	Due Date	Days	Week 11					Week 12					Week 13					Week 14				
					M	T	W	T	F	M	T	W	T	F	M	T	W	T	F	M	T	W	T	F
Development & Infrastructure Integration		3/22/2027	5/28/2027	50																				
Network Configuration	Network Infrastructure Team	4/19/2027	4/30/2027	10																				
Database Configuration	Database Administrator	5/3/2027	5/7/2027	5																				

Task & Owner		Start Date	Due Date	Days	Week 14					Week 15					Week 16					Week 17				
					M	T	W	T	F	M	T	W	T	F	M	T	W	T	F	M	T	W	T	F
Development & Infrastructure Integration		3/22/2027	5/28/2027	50																				
Application Development	Software Engineering Team	5/10/2027	5/21/2027	10																				
UWB installation and calibration on all devices	Field Techs & IT Support	5/24/2027	5/28/2027	5																				

Task & Owner		Start Date	Due Date	Days	Week 17					Week 18					Week 19					Week 20				
					M	T	W	T	F	M	T	W	T	F	M	T	W	T	F	M	T	W	T	F
Testing & Training		5/31/2027	6/22/2027	20																				
Unit Testing	Software Engineering Team	5/31/2027	6/4/2027	5																				
System Integration Testing & QA	QA Lead	6/7/2027	6/11/2027	5																				
User Acceptance Testing	Nurse Lead & QA Lead	6/14/2027	6/18/2027	5																				

Project Schedule (Cont.)

Task & Owner		Start Date	Due Date	Days	Week 20					Week 21					Week 22					Week 23				
					M	T	W	T	F	M	T	W	T	F	M	T	W	T	F	M	T	W	T	F
Testing & Training		5/31/2027	6/22/2027	20																				
Clinical Staff Training & Onboarding	Clinical Educator & Nurse Lead	6/16/2027	6/22/2027	5																				
Deployment		6/23/2027	7/7/2027	10																				
Release pilot into production	Solutions Architect & DevOps Lead	6/23/2027	6/29/2027	5																				
Post Implementation Review	Solutions Architect & Technical Manager	6/30/2027	7/7/2027	5																				

Task & Owner		Start Date	Due Date	Days	Week 21					Week 22					Week 23				
					M	T	W	T	F	M	T	W	T	F	M	T	W	T	F
Maintenance		6/23/2027	7/7/2027	10															
System Handover to IT Operations	IT Operations Lead	6/23/2027	7/7/2027	10															
UWB Tag Management	Biomedical Technicians	7/7/2027	Ongoing	Ongoing															
VLAN & Security Audits	IT Operations	7/7/2027	Ongoing	Ongoing															
Hardware & Gateway Updates	IT Operations	7/7/2027	Ongoing	Ongoing															
Calibration Management	IT Operations / Vendor	7/7/2027	Ongoing	Ongoing															
Application Management	DevOps	7/7/2027	Ongoing	Ongoing															

Validation

Accuracy Testing
Calibrate UWB tags and ceiling gateways on a targeted hospital floor to ensure each equipment's location is accurately tracked.

Network & VLAN Security Verification
Conduct security audits on the dedicated, encrypted Wi-Fi VLAN to guarantee isolation from hospital Electronic Health Records.

Nurse Search Time Benchmarking
Run timed simulations with a test group of nurses to compare manual search times against the IoT ecosystem's search speeds before full rollout.

UI/UX Usability Assessments
Conduct user acceptance testing (UAT) with floor and charge nurses to ensure maps are intuitive, as well as fast to navigate.

Stress Testing
Simulate peak network load with hundreds of active tags transmitting concurrently to verify database performance and uninterrupted server uptime.

Evaluation & Continuous Improvement

Evaluation Metrics

Efficiency & Time Savings
Track equipment lookup reductions and clinical hours returned to patient care.

Financial Return on Investment (ROI)
Quantify savings from reduced asset loss, zero duplicate purchasing, and labor efficiency gains.

System Health & Tracking Accuracy
Monitor UWB gateway network uptime, signal precision, tag battery levels, and registry validity.

User Experience & Adoption
Collect nurse feedback on dashboard UI clarity, mobile responsiveness, and map accuracy.

Infrastructure & Load Performance
Audit DB query speeds, network bandwidth usage, and peak-time map render times.

Security & Compliance
Perform periodic audits on encrypted VLAN traffic and system logs for HIPAA compliance

Continuous Improvement

Asset Redistribution
Analyze movement trends to stage equipment across units prior to peak shifts.

Workflow Refinement
Optimize UI, search filters, and inventory alert thresholds based on floor nurse input.

Proactive Maintenance
Run automated battery health checks, periodic gateway RF recalibrations, and security patches.

Training & Expansion
Refine onboarding programs and evaluate expanding UWB coverage to additional floors.

Legal, Ethical, and Cultural Considerations

Legal Considerations

- HIPAA Compliance: Strict adherence ensures location tracking is completely isolated from patient records and sensitive clinical data.
- Access & Logs: System logs and user access permissions undergo regular audits to comply with hospital policies.
- Hardware Safety: Ceiling gateway installations and UWB hardware strictly meet healthcare facility safety standards.

Ethical Considerations

- Asset-Only Focus: Real-time tracking is strictly limited to medical equipment, assuring staff the system is built purely for availability.
- Equitable Access: Location data is shared equally across all hospital units to eliminate equipment hoarding and ensure balanced patient care.
- Data Integrity: RBAC enforces strict authorization so only designated leads modify system settings or tag registries.

Cultural Considerations

- Staff Empowerment: Positioned as a supportive productivity tool to eliminate nurse search frustration rather than a staff monitoring mechanism.
- Shared Ownership: Fosters a culture of collaborative asset stewardship across shift handoffs, nursing teams, and clinical departments.

Incorporation of Feedback

Peer Feedback: Anderson St James

"One area I would consider improving is narrowing the final problem statement. It currently combines equipment tracking and patient vital monitoring into one problem..."

Implementation

- Scope Isolation: Excluded vital monitoring to eliminate scope creep and implementation hurdles.
- Targeted Focus: Directs 100% of system design toward solving equipment search times to improve critical operational bottlenecks.

Peer Feedback: Christopher Hellmuth

"Who will be taking care of the UWB tags once they are installed... break it down by what each team would be responsible for.“

Implementation

Added Maintenance Phase: Incorporated long-term support directly into the Work Breakdown Structure (WBS) and Project Schedule.

Role Assignments:

- Biomedical Technicians: Tag lifecycle, 3–5-year battery swaps, and physical hardware.
- IT Operations & DevOps: Wi-Fi VLAN security audits, gateway firmware updates, dashboard uptime, and threshold management.
- Vendor Support: Partnered with IT Operations for annual precision recalibration of ceiling gateways.

Career Readiness

Systems & Data Architecture

- IoT Hardware: Engineered a UWB sensor and ceiling gateway tracking pipeline.
- Hybrid Database Design: Structured on-premises Relational/Non-Relational schema for real-time positioning logs and asset staff assignment paring.

Enterprise Network & Security

- Network Isolation: Designed encrypted, segmented Wi-Fi VLANs to isolate tracking traffic.
- Compliance & Governance: Embedded HIPAA standards, RBAC controls, and security auditing into core system design.

Systems Engineering & Analytics

- Project Management: Built a 23-week Project Schedule and Work Breakdown Structure (WBS) for cross-functional deployment.
- Business ROI Modeling: Applied quantitative decision frameworks to project ROI, track productivity recovery, and model capital costs.

Strategic Planning & Career Application

- Holistic System Design: Evaluated legal, ethical, and workplace cultural factors to protect privacy and support equity.
- Target Roles: Serves as a production-grade project demonstrating full lifecycle systems engineering for Software Engineering and Systems Architecture roles.

Conclusion

Operational Efficiency & Patient Care

Eliminating up to 60 minutes of daily equipment search time per nurse per shift directly translates into higher quality direct bedside care and increased nursing productivity (24x7, 2024).

Financial Optimization

Implementing continuous real-time asset visibility could increase equipment utilization from 35% to 70%, preventing redundant equipment purchases and eliminating asset loss via geofencing alerts.

Enterprise Security & HIPAA Compliance

Deploying on an isolated, encrypted local Wi-Fi VLAN ensures robust security that minimizes interactions with sensitive Electronic Health Records (EHR) and Protected Health Information (PHI).

Sustainable Operations & Clear Ownership

Establishing explicit post-deployment ownership across IT Operations, Biomedical Technicians, and Vendors guarantees hardware battery lifecycles, network stability, and long-term system calibration.

Implementation Readiness

The recommended UWB IoT ecosystem is fully structured across a 23-week WBS implementation plan, providing Cook Children’s with a scalable, highly effective asset-tracking model.

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